MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 02** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Marks are indicated against each question.
2. **Section A** consists of 5 questions of 1 mark each (4 MCQs and 1 Assertion-Reason).
3. **Section B** consists of 4 questions of 11 marks total (1 VSAQ of 2 marks and 3 SA of 3 marks each).
4. **Section C** consists of 1 Case-Based integrated unit of assessment of 4 marks with sub-parts.
5. Use of calculators is strictly prohibited. Neat geometric sketches must accompany geometric proofs.

SECTION A: MULTIPLE CHOICE QUESTIONS (1 Mark Each)

1. If in two triangles ABC and DEF, $\frac{AB}{DE} = \frac{BC}{FD}$, then they will be similar when:

(a) $\angle B = \angle E$ (b) $\angle A = \angle D$ (c) $\angle B = \angle D$ (d) $\angle A = \angle F$

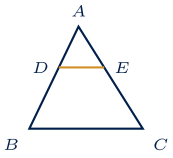
2. If $\Delta ABC \sim \Delta EDF$ and ΔABC is not similar to ΔDEF, then which of the following is **NOT** true?

(a) $BC \cdot EF = AC \cdot FD$ (b) $AB \cdot EF = AC \cdot DE$ (c) $BC \cdot DE = AB \cdot EF$ (d) $BC \cdot DE = AB \cdot FD$

3. If $\Delta ABC \sim \Delta PQR$ with $\frac{BC}{QR} = \frac{1}{3}$, then the ratio $\frac{\text{Perimeter}(\Delta PQR)}{\text{Perimeter}(\Delta ABC)}$ is:

(a) 9 (b) 3 (c) $\frac{1}{3}$ (d) $\frac{1}{9}$

4. In ΔABC, D and E are points on sides AB and AC such that $DE \parallel BC$. If $AD = 2$ cm, $DB = 3$ cm, and $DE = 4$ cm, then the length of BC is:



(a) 6 cm (b) 8 cm (c) 10 cm (d) 12 cm

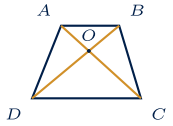
5. **Assertion (A):** All equilateral triangles are similar to each other.
Reason (R): All equilateral triangles have each interior angle equal to 60°, satisfying the AAA similarity criterion.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

SECTION B: DESCRIPTIVE QUESTIONS (PART 1)

6. **[2 Marks]** In ΔABC, $DE \parallel BC$ such that $AD = 4$ cm, $DB = (x - 4)$ cm, $AE = 8$ cm, and $EC = (3x - 19)$ cm. Find the value of x.

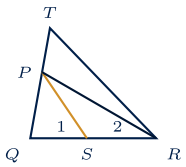
7. **[3 Marks]** The diagonals of a quadrilateral $ABCD$ intersect each other at the point O such that $\frac{AO}{BO} = \frac{CO}{DO}$. Show that $ABCD$ is a trapezium.



SECTION B: DESCRIPTIVE QUESTIONS (PART 2 Continued)

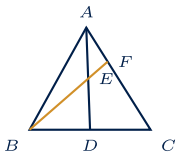
8. **[3 Marks]** In the given figure, $\frac{QT}{PR} = \frac{QR}{QS}$ and $\angle 1 = \angle 2$. Prove mathematically that:

$$\Delta PQS \sim \Delta TQR$$



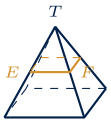
9. **[3 Marks]** In ΔABC, D is the midpoint of BC and E is the midpoint of AD. If BE produced meets AC at F, prove that:

$$AF = \frac{1}{3}AC$$



SECTION C: CASE STUDY BASED QUESTION (4 Marks)

10. **Case Study 02: Architectural Scale Model & Farmhouse Roof Pyramid**



A student prepares a scale model of a farmhouse roof for an exhibition. The attic floor, $ABCD$, is a square of side 12 m. The roof beams form a pyramid with apex T, where each lateral edge AT, BT, CT, DT measures 12 m. Supporting horizontal beams form a square $EFGH$ where E, F, G, H are the midpoints of AT, BT, CT, DT respectively.

(i) **[1 Mark]** State the similarity criterion between ΔTEF and ΔTAB.

(ii) **[1 Mark]** Find the length of the horizontal supporting beam EF.

(iii) **[2 Marks]** What is the ratio of the perimeter of square $EFGH$ to the perimeter of square $ABCD$? Justify your answer.

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